HridayMitra: catching worsening heart failure at home, 2 days before the hospital

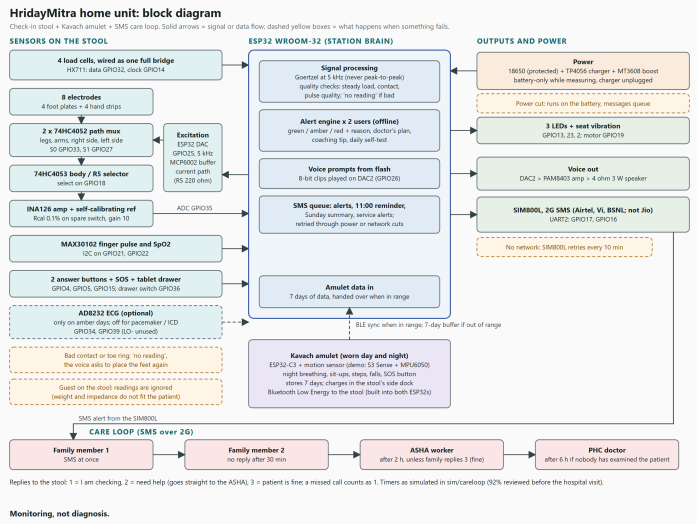
Transfinitte hardware track · Dare 2 Dream Ventures PS · **Monitoring, not diagnosis.** All patient data synthetic; no human data collected.

Problem

Heart-failure patients are re-admitted because fluid builds up silently for days. Daily weighing alone catches about one in three events and floods families with false alarms; sticky patches are not worn by elders; rural homes have power cuts, 2G and low literacy.

Solution: a 90-second check-in station + an amulet + a care loop

- **Check-in stool** (₹4,367 build cost): weight (4 load cells), **8-electrode bioimpedance spectroscopy** (5/20/50 kHz: water outside cells) (legs, arms, both sides → chest fluid by segmental subtraction), finger pulse + SpO_2, stand-up time, 3 spoken yes/no questions answered with big buttons.
- **Kavach amulet** (ESP32-C3 + accelerometer, breakaway cord, no magnets): night breathing rate, night sit-ups (orthopnea), activity, falls, SOS → BLE to the stool.
- **Offline alert engine on the ESP32**: learns each person's own normal, scores only the worsening direction, needs **two independent witnesses** (fluid, heart-lungs, daily life) before red, and says **why** in plain words.
- **Closed care loop over SMS**: family 1 → family 2 (30 min) → ASHA (2 h) → PHC doctor (6 h); reply 1/2/3 or a missed call.
- **A product, not a gadget**: speaks the **doctor's own sick-day plan**; coaching tips (salt, missed tablets, infection-like); **two users per stool** with separate baselines; tablet drawer that knows if medicine was taken; seat vibration for hard-of-hearing users; daily **self-test** with "needs service" SMS; 11:00 missed check-in reminder; Sunday summary to the ASHA; village dashboard; OPD report and **ABDM/FHIR export**.



User journey (≈ 90 s a day)

Sit → voice: "feet on the plates, hold the handles" → finger on the light → 3 questions (red = yes, green = no) → stand up → light + spoken result. Red: family gets the SMS with reasons, the ASHA is told at 2 h unless the family replies "3 fine".

Results (frozen threshold tuned on 140 training patients, tested once on held-out patients)

	HridayMitra engine	Weight rule (SCALE-HF)	Logistic regression (ML)
Sensitivity, held-out (35 events)	**66%**	40%	80%
Specificity	98.7%	97.6%	-
Median lead before hospital visit	3 days (amber heads-up: 4 days)	2 days	2 days
False alerts per patient-month	**0.06**	0.53	0.12
External 1,000-patient cohort: sensitivity / false alerts	**72% / 0.06**	31% / 0.38	81% / 0.08

Data: 200 pretend patients × 60 days (PS cohort, 60% with a visit) + 1,000 × 180 days (10%), latent-congestion model with salty meals, infection, missed pills, guests, wet feet, missing days; calibrated so the weight rule reproduces its published sensitivity (31% vs 35%) and false-alert rate (0.38 vs 0.34). Patient-level 70/30 split, seed 42; no-peeking check on every system.

Robustness: we tried to break it

Test	Result
Ablations (remove one sensor group)	no impedance 65%, no weight 66%, no amulet 54% (full engine 72%): every group earns its place
Stress: 50% missing days / guests / wet feet / power cuts / electrode ageing / everything at once	47% … 57% sensitivity; 10% guests: engine 0.10 false alerts vs weight rule **3.2**
Assumption sweep: each of 23 guesses pushed low and high (52 new cohorts)	beats the weight rule in **52/52**, inside the false-alert budget in 52/52; sensitivity 30%–98%, driven by how fast worsening develops, not by hardware noise
Care-loop simulation (5,000 alerts per scenario)	clinical review before the hospital visit: **92%** with the ladder vs 27% with one SMS; median 6 h; everything bad at once 68%
Heart rate on real ECG (PhysioNet BIDMC CHF, 15 records)	median heart-rate error 0.15 bpm; 13 of 15 records within 0.5 bpm; 96% of 30-s windows within 5 bpm; fails on chf12 (very noisy recording)

Hardware and simulation proof (free tools: LTspice, KiCad, Python)

- Tetrapolar, ratiometric front end ($Z = R5 \cdot V_{body}/V_{R5}$): gain and contact resistance cancel; LTspice error ≤ 0.05% for contacts 100 Ω–10 kΩ; +10% leg fluid → −8.7% Z; 134 µA excitation.
- **Monte Carlo, 100 stools from 5% parts** on random bodies and skin: 99% of readings within 1.3%, a +10% fluid change measured within 0.10 points. It caught a design flaw: 1 MΩ bias resistors gave 12% errors → **changed to 10 MΩ**.
- **Bioimpedance spectroscopy (5/20/50 kHz, Cole fit on the chip)**: +10% fluid outside the cells moves the water index +3.0%; foot position / temperature (+3%) moves the single-frequency reading +3.0% but the index +0.0%; fluid inside cells moves it the other way (-2.9%). Engine v3 with it: external-cohort sensitivity 72% at 0.06 false alerts.
- **Patient safety (IEC 60601-1, type BF)**: every single fault stays within the patient-current limits (≥ 12.8× margin); a CHARGE / USE switch cuts charger leakage from 339 to 0.19 µA.
- **Self-calibrating reference** (0.1% Rcal on the spare switch, checked every check-in): absolute error median 0.16%; a damaged reference is caught and reported for service.
- Mains hum (230 V coupling, charger plugged in): Goertzel reading ≤ 0.16% error vs 18% for peak-to-peak → firmware uses Goertzel only.
- KiCad schematic generated from code: ERC 0 violations; the netlist is checked against the path table and the LTspice values automatically.
- Engine ported to C++ for the ESP32: **100% identical lights** to the Python engine over 3,231 patient-days.
- **Software-in-the-loop**: the real firmware runs a scripted month in an emulated world (ADC body signal, HX711, MAX30102, SIM800L, BLE, flash): 26/26 checks pass (care loop, reminders, self-test, power cut, network outage, two users) and it found 3 bugs before any board was flashed.

Edge cases handled

Situation	What happens
Guest sits on the stool	weight jump > 2.5 kg → readings ignored
Toe ring / bad foot placement	leg reading > 25% off → "no reading", voice asks to re-place feet
Power cut, no network	18650 battery; SMS queued, retried every 10 min
Missed days	network clock fills gaps; light held
Infection vs fluid	cause tag: pulse/breathing up with steady fluid = infection-like
Oxygen below 90%	red at once, everyone messaged
Low literacy	voice, 3 lights, 2 buttons, missed call = reply
Pacemaker / ICD	no magnets; impedance and ECG off by setting

Cost and scale

Build cost ₹4,290–4,445 (Monte Carlo of the itemised list; limit ₹5,000); ≈ ₹2,620–3,057 at 100 units. SMS ≈ ₹0.05 per patient-month. Fits ASHA-led programmes: one stool per home, the ASHA card closes the loop.

Honest limits

- Synthetic data: shows the method works on modelled patterns, not clinical accuracy. Lead time is bounded by the PS worsening window (3–10 days, accelerating).
- Chest impedance is weak (needs ~7 days of averaging); circuit not yet compared with a bench build; SpO_2 calibration generic; ML regression catches more events but with 3–5× the false alarms.
- Never for anyone with a pacemaker or ICD in the impedance mode; battery only during measurement. Code, data, tests: github.com/aadityaa0523/HridayMitra